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BANGALORE

A Project Report

On

"Finance Receipting using Robotic Process Automation"

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1. INTRODUCTION

Our problem statement revolves around automating a back-office process for generating and sending receipts based on transaction details from 25,000 receipts per month.

Back-office operations typically involve a range of administrative and support tasks that are essential for the smooth functioning of a business. These tasks are often repetitive, structured, and rule-based, making them ideal candidates for automation. In the current problem, the focus is on automating the processing of structured data, segregating clients based on transaction types, generating receipts using predefined templates, and sending them to respective vendors.

With the advent of Robotic Process Automation (RPA) and other automation tools, businesses can automate such manual, high-volume tasks, reducing errors, increasing efficiency, and freeing up human resources for more complex tasks. In this case, the process involves handling different receipt types (Card/Cheque), using pre-defined templates, and generating outputs based on specific rules and conditions.

Automation technologies like Robotic Process Automation (RPA) offer powerful solutions for processes like these. RPA is especially suited for high-volume, repetitive tasks that require interaction with multiple systems, such as downloading data from SAP, segregating it based on transaction types, and feeding it into a web application. In this case, the client's process also involves generating receipts using transaction-specific templates stored on a file server and sending the receipts via email to respective vendors. Automating these steps ensures that transactions are processed more quickly and accurately while reducing human involvement in mundane tasks.

One of the key challenges in this automation process is dealing with exceptions, which account for 10% of the cases. Exception handling in automation involves designing intelligent workflows that can manage irregular data or transaction types without breaking the system. Another challenge is integrating automation with existing systems like SAP and a web application, ensuring smooth data flow and timely execution. A well-designed automation solution can help the client streamline this entire workflow, improving speed, accuracy, and overall operational efficiency.

2.LITERATURE REVIEW

METHOD OVERVIEW:

1. The current state, trends, and implementation of Robotic Process Automation (RPA) across both scientific research and industrial applications.

Advantages:

RPA offers cost reduction, improved efficiency, non-invasive system integration, better compliance, enhanced employee satisfaction, and consistent service quality through automation of repetitive tasks.

Limitations:

RPA faces challenges with unstructured data handling, interface dependencies, high setup costs, maintenance needs, limited scope to rule-based processes, and requires extensive training and process documentation.

2. The implementation and effectiveness of digital receipts in online transactions, focusing on their role in the reconciliation process and financial report preparation.

Advantages:

Digital receipts streamline the reconciliation process, reduce manual errors, enable faster financial report preparation, and provide better transaction tracking and documentation in online business operations.

Limitations:

Digital receipt systems face challenges with standardization across different platforms, security concerns, data storage requirements, and potential technical issues during system integration or network disruptions.

3. Adoption patterns, factors, and implementation strategies of Robotic Process Automation (RPA) across organizations and industries.

Advantages:

RPA adoption brings operational efficiency, cost reduction, improved accuracy, increased employee satisfaction, better compliance, and scalability of business processes without major infrastructure changes.

Limitations:

Organizations face challenges in RPA adoption including high initial costs, resistance to change, limited capability for complex tasks, technical integration issues, and the need for continuous maintenance and skilled personnel.

4. Impact of Robotic Process Automation (RPA) in shaping the future digital workforce

Advantages:

RPA enhances workforce productivity by automating repetitive tasks, reducing operational costs, improving accuracy, enabling 24/7 operations, and allowing human workers to focus on more strategic and value-added activities.

Limitations:

The implementation of RPA poses challenges related to job displacement concerns, requires significant investment in training and infrastructure, and faces technical limitations in handling complex, judgment-based tasks and unstructured data.

5. Implementation and impact of Robotic Process Automation (RPA) specifically in

accounting and auditing processes

Advantages:

RPA in accounting and auditing enhances accuracy, speeds up financial processing, reduces human errors, improves compliance, ensures consistent reporting, and enables real-time financial data analysis and monitoring.

Limitations:

RPA implementation in accounting/auditing faces challenges with complex financial decision-making, requires substantial initial investment, needs continuous updates for changing regulations, and may struggle with non-standardized financial documents and unconventional transactions.

6. Transformative role of Robotic Process Automation (RPA) in modern accounting practices

Advantages:

RPA modernizes accounting by automating routine tasks, improving data accuracy, enhancing compliance, reducing processing time, enabling real-time reporting, and freeing accountants to focus on strategic analysis and decision-making activities.

Limitations:

The implementation of RPA in accounting faces barriers including high initial costs, resistance to change from traditional practices, complexity in handling non-standardized accounting processes, and the need for continuous system updates and staff training.

→EXISTING METHODS:

Advantages of Existing Methods

1. (Manual Processing): Low Initial Investment
   Manual Processing: Requires minimal technology investment, as it relies on human labor. This makes it cost-effective in the short term, especially for businesses with low volume.

2. (Manual and Semi-Automated): Flexibility in Handling Complex Cases
   Humans can easily adapt to changes in rules, regulations, or exceptions in complex scenarios, while semi-automated systems allow for some level of manual intervention when needed.

3. (Manual Processing): Ease of Implementation
   Requires no specialized training or technology skills to set up. Processes are easy to start with existing tools (Excel, email clients), so businesses can implement without major changes to their infrastructure.

4. (Manual and Semi-Automated):Control over the Process
   Manual processes allow for more control and direct oversight, meaning each stage can be reviewed by humans before the next action is taken. This is particularly useful in businesses that prioritize accuracy over speed.

5. Ability to Handle Physical Documents (Manual):

- Paper receipts and documents are handled naturally in a manual process. In semi-

automated systems, basic OCR tools may be used to extract data from physical documents.

6. (ERP Methods):ERP System Familiarity
   Businesses that already use ERP systems (like SAP) are familiar with their interfaces and basic workflows. This familiarity can reduce the learning curve for employees when managing processes like receipt generation.

7. (ERP Methods):Transactional Visibility
   ERP systems like SAP offer transaction logs and audit trails that provide detailed tracking of each transaction. This can help with compliance and reporting requirements, although it often requires manual intervention.

Limitations (Disadvantages) of Existing Methods

1. (Manual Processing): Time-Consuming
   Processing 25,000 receipts manually is highly time-intensive. Employees must segregate data, choose templates, send emails, and handle exceptions, leading to slower response times and inefficiency.

2. (Manual and Semi-Automated):High Error Rate
   Manual entry or semi-automated scripts are prone to human error, especially when dealing with large volumes. Mistakes in data entry, template selection, or email addresses can cause delays and inaccuracies.

3. (Manual and Semi-Automated): Scalability Issues
   Manual or semi-automated methods do not scale efficiently. As the volume of receipts increases, the process becomes slower and requires more staff, leading to higher costs and processing delays.

4. (Manual and Semi-Automated):Lack of Real-Time Processing
   Manual and basic automation methods can’t handle real-time data retrieval from SAP or immediate email dispatch. This delay affects business agility and the ability to respond quickly to changing requirements.

5. (Manual and Semi-Automated): Dependency on Skilled Labor
   Highly manual processes are dependent on skilled labor. Employee turnover, absence, or fatigue can disrupt operations, leading to inconsistent performance and potential delays.

6. (Manual, Semi-Automated): Integration Challenges
   Manual or semi-automated processes often require data to be manually imported or exported between systems (e.g., SAP, email clients, file servers). This makes integration inefficient and prone to errors.

7. (Manual, Semi-Automated): Lack of Real-Time Monitoring
   o Manual and semi-automated processes lack real-time tracking and reporting capabilities. Businesses must rely on post-process reporting or audits, which do not offer immediate insights into issues like exceptions or errors.

3. OBJECTIVES

- Primary Objective:
  To design and implement a Robotic Process Automation (RPA) solution that automates the finance receipting process, handling 25,000 receipts per month efficiently and accurately.

- Develop an automated system to download and process structured data from the SAP application.

- Implement logic to segregate client details based on transaction type (Card or Cheque).

- Design a solution to integrate paper-based receipts into the automated workflow, possibly using Optical Character Recognition (OCR) technology.

- Maintain high accuracy in data processing and receipt generation, ensuring compliance with financial regulations and company policies.

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4.EXPERIMENTAL DETAILS/METHDOLOGY

Software Tools:

- RPA Tools: UiPath, Automation Anywhere, Blue Prism (for RPA-based solutions).
- Programming Tools: Python, JavaScript (for custom scripting).
- IDE: Visual Studio Code, PyCharm (for development).
- Database: SQLite, (for data storage).
- Email Automation: Gmail/Outlook SMTP (for email sending).
- Document Generation: Python-docx, Microsoft Word (for receipts/reports).
- Web Automation: Selenium, Puppeteer (for web-based automation).
- Cloud Platforms: Heroku, AWS Lambda (for deployment in the cloud).
- Version Control: GitHub, GitLab (for collaboration).

Technology Stack Components:

- Robotic Process Automation (RPA).
- SAP Integration.
- Data Management & Storage.
- Data Segregation and Rules Engine.
- Document Generation (Template-Based).
- Email Automation.
- Exception Handling.
- Optical Character Recognition (OCR) (for handling paper-based inputs).
- Security & Compliance.
- Reporting and Monitoring.
- Collaboration & Version Control.

5.METHODOLOGY

Methodology for Finance Receipting Automation using RPA

To automate this back-office process efficiently, the following methodology will be used:

1. Requirements Gathering:
   - Identify the current process steps.
   - Understand the data structure in SAP.
   - Identify exceptions and how they are handled.
   - Establish security and compliance requirements.
2. Process Mapping:
   - Map the manual process of downloading, segregating, and generating receipts.
   - Identify key decision points for rule-based segregation.
   - Design exception handling flows.
3. Technology Stack Identification:
   - Robotic Process Automation (RPA) for automating rule-based actions.
   - Optical Character Recognition (OCR) for paper-based receipts.
   - SAP Integration for extracting data from the SAP application.
   - Document Generation Tool for template-based receipts.
   - Email Automation Tool for sending receipts to vendors.
   - Monitoring and Reporting Dashboard for tracking performance and exceptions.
   - Security and Compliance features for data privacy and access control.
4. Development and Testing:
   - Build the RPA bots to automate the process.
   - Test the automation in a simulated environment using various input data.
   - Handle exception scenarios with manual and automated methods.
5. Deployment and Maintenance:
   - Deploy the RPA solution in the production environment.
   - Monitor the performance and exceptions using the dashboard.
   - Ensure ongoing maintenance and updates to accommodate any changes in templates, rules, or regulations.

MODULES

1. SAP Integration Module:
   - Function: Automatically extracts receipt and client details from SAP.
   - Technology: SAP API or connector.
   - Output: Structured data (transactions, client details).
2. Data Segregation and Rules Engine:
   - Function: Segregates transactions based on type (Card or Cheque).
   - Technology: RPA workflow logic.
   - Output: Segregated data, flagged exceptions (10%).
3. Document Generation Module:
   - Function: Generates receipts based on transaction type using predefined templates.
   - Technology: RPA or document automation software (e.g., DocuSign, MS Word automation).
   - Output: Generated receipts (PDFs).
4. Email Automation Module:
   - Function: Sends generated receipts to corresponding vendors.

- Technology: SMTP or email automation tools (e.g., Outlook API, Gmail API).
- Output: Sent email confirmation.

5. OCR and Exception Handling Module:
    - Function: Handles paper-based inputs via OCR and manages flagged exceptions.
    - Technology: OCR tools (e.g., ABBYY, Tesseract), RPA exception handling workflows.
    - Output: Parsed data from paper receipts and exception reports.
6. Monitoring and Deployment Module:
    - Function: Tracks system performance, monitors exceptions, and generates reports.
    - Technology: RPA dashboard, SQL database for logs, visualization tools
    - Output: Reporting and deployment of automation on web application

5. OUTCOMES

1. Increased Efficiency and Reduced Manual Work:
- Automation of routine tasks such as data extraction, segregation, and email generation will significantly reduce manual work.
- The automated process will handle 90% of the receipts without human intervention, increasing overall efficiency.

2. Reduced Errors and Improved Accuracy:
- Automation ensures fewer errors in tasks like data entry, document generation, and email distribution.
- Exception handling for 10% of cases ensures a streamlined process with minimal manual correction.

3. Faster Processing Time:
- The entire process, from data extraction to receipt generation and distribution, will be faster due to automation of repetitive tasks, leading to quicker turnaround times.

4. Scalability:
- The system will be scalable to handle more than 25,000 receipts per month as business needs grow, without adding extra resources.

5. Cost Savings:
- By reducing manual labor and the need for human oversight in routine tasks, there will be cost savings in terms of personnel and time spent on these activities.

6. Improved Compliance and Security:
- With secure email automation and the inclusion of data validation checks, the system will ensure compliance with regulatory requirements related to finance and data security.
- Audit trails and reports will be generated for monitoring and compliance purposes.

7. Error Tracking and Exception Handling:
- The system will identify and highlight exception cases (10%) that need manual review, reducing error propagation and ensuring human oversight where necessary.

8. Better Customer/Vendor Relations:
- Automated and timely email delivery of receipts to vendors will improve vendor communication and relations, enhancing professionalism and accuracy in client-vendor interactions.

9. Streamlined Documentation and Workflow:
- Document generation based on transaction types will ensure standardized

6. TIMELINE OF THE PROJECT/ PROJECT EXECUTION PLAN

1. Planning And Requirements (12-Sep-2024 To 18-Sep-2024)

- Define project scope, objectives, and outcomes.
- Identify limitations and gather requirements.
- Develop project timeline and resource allocation.

2. Design and Prototyping (15-Oct-2024 To 21-Oct-2024)

- Analyze data sources (SAP data, paper-based receipts).
- Design data flow and process flow diagrams.
- Define rules for data segregation (Card/Cheque) and exception handling.
- Identify key templates for document generation.
- Design architecture for OCR integration and email automation.

3. Development (21-Oct-2024 To 19-Nov-2024)

- Set up the RPA environment and integration with SAP.
- Implement OCR for paper-based inputs.
- Develop automation scripts for data segregation, receipt generation, and email distribution.
- Implement document generation and file server integration.
- Build exception handling workflows.
- Develop reporting and monitoring dashboard.

4. Testing Phase (21-Oct-2024 To 19-Nov-2024)

- Unit testing of individual modules (data extraction, segregation, document generation).
- Integration testing for full end-to-end automation.
- Validate exception handling and test with sample edge cases.

5. Deployment (19-Nov-2024 To 17-Dec-2024)

- Finalize deployment strategy and deploy the solution to production.
- Provide necessary training and documentation for users.
- Monitor initial execution for any post-deployment issues.
- Rollout email automation for all vendors.

6. Evaluation, and Optimization (19-Nov-2024 To 17-Dec-2024)

- Evaluate system efficiency and identify areas for improvement.
- Address any outstanding issues or performance bottlenecks.
- Fine-tune reporting and dashboard elements for enhanced visibility.

7. CONCLUSION

In this project, we have designed and will implement an automated Finance Receipting solution using Robotic Process Automation (RPA) to streamline and optimize back-office operations. The automation solution is capable of handling a high volume of 25,000 receipts per month, with minimal manual intervention, while addressing the challenges posed by structured and paper-based data inputs. By leveraging RPA to automate tasks such as data extraction, segregation, receipt generation, and email distribution, we have successfully enhanced operational efficiency, accuracy, and scalability.

The proposed solution will effectively reduce human errors, improve the speed of processing, and ensured compliance with business rules, while also providing an exception handling mechanism to address outliers (10% of cases). This project demonstrated the significant impact of automation on reducing repetitive manual tasks, improving cost-efficiency, and enhancing overall productivity. The inclusion of a real-time reporting and monitoring dashboard provided transparency, enabling better decision-making and process optimization.

In conclusion, the automated finance receipting system presents a scalable, robust, and efficient approach to managing high-volume financial transactions, which can be easily adapted to other industries with similar repetitive tasks. The solution not only addresses current business needs but also sets the stage for future advancements in automation and process optimization, offering considerable long-term benefits for organizations.

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